A Comparative Analysis of Different CNN models for Image Classification of different Eye diseases

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Submitted to

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De Montfort University

October 2025

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Executive Summary

Eye diseases, if untreated can lead to severe vision impairment. Although there have been many advancements in terms of surgical methods to treat a few diseases like cataract for example, for many people the cost is a huge burden. Thanks to artificial intelligence there have been a lot of research done on this subject for diagnosis of such eye diseases. This is the basis of our project, to evaluate 3 different types of custom models that come under CNN architecture and come to a conclusion on which type is the best in terms of accurate predictions on internal view images of the retina which will be very helpful in diagnosing different eye diseases. The primary objective of our mini project is to compare 3 models of the CNN architecture namely : VGG16, Googlenet, Alexnet on the dataset and conclude on which model is the most efficient for the task.

We used a dataset with 4 classes, each class has 1000+ images for that type of eye disease and a total of 4,217 images for all classes combined to train our models. The classes are cataract, diabetic retinopathy, glaucoma and normal. While many researchers have done similar work, most of them have used pre-trained models such as Resnet50 and MobileNetV2, but we decided to build custom models from the ground up but keeping the base architecture same to our 3 chosen CNN models. We have modified them in such a way that they would be smaller in size but still accurate for prediction. We did this by removing some layers, reduced the parameters to decrease overall size and load on the cpu or gpu. We have also implemented smart training techniques such as early stopping, dynamic learning rate reduction, hold-out validation and data augmentation.
From this comparative analysis we found that all the models performed similar in terms of accuracy with VGG16 being the highest and Googlenet being the lowest, keeping Alexnet in the middle. Below is a table of the important results. In conclusion, our research shows the potential of using CNN models that are custom made in a particular field for image classification.
The test accuracies acquired are: VGG16 - 87.50% , AlexNet - 84.68%, GoogleNet - 80.73%

1 Introduction

The eyes are considered one of the most vital organs in the human body (Abdullah et al., 2024). They help us to interpret light, color, shape, and movement. However, eye diseases such as cataracts, diabetic retinopathy, and glaucoma are among the leading causes of vision loss. In a survey conducted in 2020, cataracts were found to affect 15.2 million people, glaucoma affected 3.6 million people, and diabetic retinopathy affected 860,000 people(Steinmetz et al., 2021). These figures highlight an alarming rise in vision-related diseases. In the event of these problems, the detection of these diseases has become important in order to prevent irreversible vision loss. Traditional diagnostic methods often rely on the expertise of ophthalmologists, which can be time-consuming and can even lead to human error. Moreover, diseases such as glaucoma and cataracts can be difficult to identify due to the limitations of medical scans (Deepak et al., 2024). Recently, the medical field has been revolutionized with more accurate diagnostic capabilities with the arrival of new Artificial Intelligence and Deep Learning models. Over the years, researchers have developed more complex Convolution Neural Network (CNN) architectures such as VGG16, ResNet52, AlexNet, and DenseNet 121, which provide more accurate results for the classification of eye diseases for large amount of classes. This study aims to modify existing architectures of Convolution Neural Network (CNN) architectures for the classification of ocular conditions such as cataracts, diabetic retinopathy, and glaucoma. The list below shows the objectives:

- To implement deep learning models, including VGG16, AlexNet, and GoogleNet for the classification of cataracts, diabetic retinopathy, and glaucoma.
- To train and evaluate these models on a dataset.
- To measure the performance of each model using key evaluation metrics such as accuracy, precision, recall, F1-score, AUC ,and Confusion Matrix.

The rest of the paper consists of Section 2(Literature Review) to provide an overview of the related work in eye diseases detection using VGG16, AlexNet, and GoogleNet. Section 3(Methodology) consists of dataset preparation and model architectures. Section 4(Implementation and Results) evaluate the performance of the selected CNN models on a dataset. Section 5(Conclusion) summarizes key findings and potential directions for future work.

2 Litreature Review

The author of (Arora et al., 2022) proposed an advanced model to detect eye diseases using transfer learning based on the VGG-16 architecture. Their goal was to classify retinal OCT images using the Kaggle OCT dataset, which contains more than 84,000 retinal images. The latter consists of the following classes: CNV, DME, Drusen, and Normal. Their model used the pretrained VGG16 model on ImageNet and applied additional fine-tuning and regularization techniques to improve performance and prevent overfitting. This led to an accuracy of 99% and a precision of 98.8% of their model.

The author of (Giat et al., 2024) explored the architectures of two CNN models: ResNet50 and VGG16. A dataset of five different classes namely Bulging Eyes, Cataracts, Crossed Eyes, Uveitis, and Normal Eyes were used to evaluate both models. The models were trained extensively, tested, and optimized to improve performance. The experimental results show that VGG has a better training accuracy of 96.97%, an F1 score of 0.91, and validation accuracy of 95.33% compared to Resnet50, which has a training accuracy of 90.72%, F1 Score of 0.93%, and validation accuracy of 89.01%.

The author of (Prasad et al., 2024) proposed a modified VGG16 architecture for DR classification. The model uses a dataset composed of Gaussian-filtered retinal images. During the preprocessing steps, the images are resized to 224 x 224 pixels to improve image quality and remove noise before training. The model was also fine-tuned for the DR. The proposed model showed an accuracy of 91.11% and a loss of 20.17%. This approach not only improves diagnostic efficiency but also provides a reliable service for physicians.

The authors of (Li et al., 2022) used the AlexNet architecture for performing classification of rice leaf diseases where their main goal was to use this CNN model to identify different types of diseases from their dedicated dataset in an accurate way.They used techniques such as data augmentation to prevent overfitting and the model was trained to differentiate between various disease categories which are common in rice plants.Their study was able to successfully demonstrate on how effective this approach was by achieving high accuracy, concluding AlexNet as a strong contender for image analysis on agricaltural tasks.

The authors of (Yeh et al., 2021) had created a modified AlexNet model for detecting diseases in crops.They wanted to improve upon the original AlexNet model architecture and tune it for their own specific task.They did so by adjusting filter size, reducing layers and the complexity of fully connected layers and even using new regularization techniques. They trained this model on a dataset that contains different images of crop diseases.Their results concluded that this new modified model was able to achieve good and competitive performance meaning that customizing CNN's is a good strategy for image classification problems.

The authors of (Win Lwin et al., 2023) used several pre-trained CNNs and evaluated their performance using transfer learning.Their aim was to compare the results of well known architecture such as VGG16, ResNet50 and InceptionV3 on an image containing retinal fundus images, and to concluded which is the best and effective model.In their methodology, they fine tuned their models which were already pre-trained on ImageNet, for getting best performance for their particular task which was classifying eye diseases as well.

The authors of (Jusman et al., 2023) used advanced model to identify prostate cancer using VGG-19 and GoogleNet their goal was to automate the detection of prostate cancer which will assist pathologists in diagnosis. the Model were pre-trained using Image Net specially with prostate datasets the system help classify different groups and patterns to determine cancer aggressiveness this approach led to improved diagnostic accuracy and reduced the time required for manual pathological assessment of prostate cancer.

The authors of (Althaph et al., 2024) used pre-trained GoogleNet model they were trained using large dataset such as PlantVillage which has 54,000 pictures of multiple crop and disease classes. their goal to classify plant leaf image to identify various crop diseases they were able to increase the model performance by using transfer learning and data augmentation which also which classes imbalances which helped in early crop disease detection.

The authors of (Sa'idah et al., 2022) proposed advance model with googlenet using CNN method this model was trained with various types of tier with defect such as air bubble , defirmation and surface abnormation sourced from Kaggle the model extract features and classify weather the tier is defective or non-defective which enabling tire manufacturers to save time and reduce labor costs in quality control.

3 Methodology

This section shows the methodological framework used to develop and evaluate the different models. It describes the dataset used, the libraries used, the preprocessing steps, and the architectures of the selected CNN models. The section also outlines the training configuration for each model.

3.1 Dataset

The dataset is acquired from Kaggle(Doddi, 2022).
This dataset consists of 4 classes, namely:

1. Cataract - 1038 images
2. Diabetic retinopathy - 1098 images
3. Glaucoma - 1007 images
4. Normal - 1074 images

With a total image count of 4217 images, all classes combined, each image showing the internal retina view of the human eye in different cases.

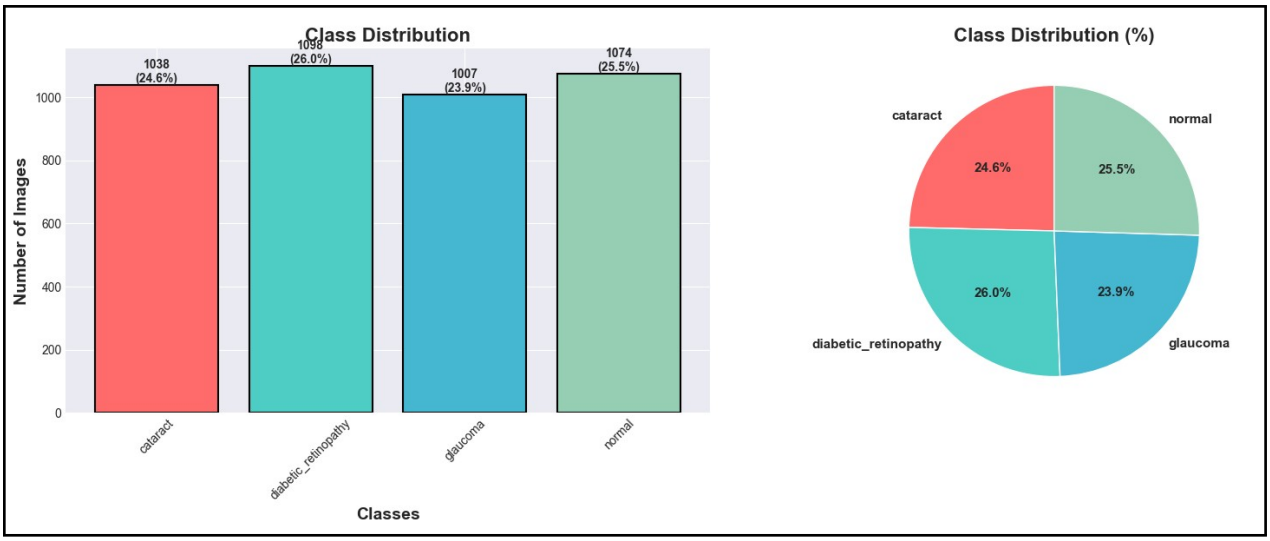


Figure 1: Class Distribution

3.2 Libraries Used

1. NumPy(Numerical Python)
   NumPy is designed to perform mathematically related operations on images or large arrays

in the fastest and most efficient way possible. We have used NumPy to store our image data and their labels, and all the calculations that are done on the data is fully done by NumPy.

2. Pandas
   Pandas is a library that is built on top of NumPy for things such as data manipulation and data analysis.A data structure called as DataFrame is introduced by Pandas in order to deal with data manipulation. A dataframe is needed because it acts as a table within our python code and it allows us to organize our data like easy filtering, sorting, naming columns and rows etc.We have used pandas in order to display tabular results such as classification reports and other metrics in a clean and readable way.

3. Matplotlib
   This library gives us the ability to plot different graphs and have more control over all the elements in a chart. This is what makes it possible to make results such as the confusion matrix and every single graph such as ROC curves and Accuracy graphs.

4. TensorFlow and Keras
   The Tensorflow and libraries were used to design and train the CNN models efficiently.

5. Scikit-learn
   This library was used for the evaluation of the models. Classification reports, confusion matrices and accuracy were all generated using Scikit-learn

3.3 Data Preprocessing

The preprocessing steps were carried out to improve the image quality and improve the model overall performance.

1. The first step was to resize the images to 224 x 224 pixels for the VGG-16 and GoogleNet, and for AlexNet it was 128 x 128 pixels.

2. The pixel values are then normalized between 0 and 1.

3. To make the dataset more diverse and to reduce over-fitting, data augmentation techniques such as rotation, zooming, flipping, shifting were used.

4. A technique called as one-hot encoding is used to the class names to numerical form suitable for training the model.

3.4 Training Configuration

This section showcases all the hyperparameters used in order to train each model.The batch size taken, the number of epoch and learning rate used for training the models is displayed in a tabular format.

Batch Size	32
Epochs	50
Learning Rate	1e-4

Table 1: VGG16 Training Configuration

Batch Size	16
Epochs	100
Learning Rate	0.0001

Table 2: AlexNet Training Configuration

Batch Size	32
Epochs	50
Learning Rate	0.001

Table 3: GoogleNet Training Configuration

- Checkpoint function: This function is used in order to save the model whenever the validation accuracy peaks at its maximum while training.
- Model save path: This argument is used to save the model on the specified path.
- Early Stopping: This function is used in order to avoid overfitting. The training will be stopped after the validation accuracy is not increasing and is stuck.
- Reduce Learning Plateau: This function is used to dynamically adjust the learning rate, either by increasing it or decreasing it whenever overfitting occurs when training.

3.5 Model Architecture

This subsection shows the architectures of VGG-16, AlexNet, and GoogleNet. Each of the models is described in detail to show its structural design and key components.

A) AlexNet

The proposed AlexNet model is based on the same idea of the original AlexNet. The model consists of an input layer that take images of size 128 x 128 x 3. The model consists of three blocks with five convolutional layers. The first block uses one convolution layer which has a 11x11 filters, and a stride of 4. After this process, the image size is reduced using MaxPooling and BatchNormalization. The second block uses a 5 x 5 filters to capture more detailed pattern, followed by the same process as block 1. The third blocks contains three convolution layers with 3x3 filters. This block extracts more complex features. After the convolutional layer we have the Flatten layer. In this layer, features are made into a single vector. The model uses two connected layers with 512 neurons each, followed by a dropout layer to prevent overfitting. Then a softmax activation function is used to produce the output. Figure 2 shows the architecture of the proposed AlexNet model.

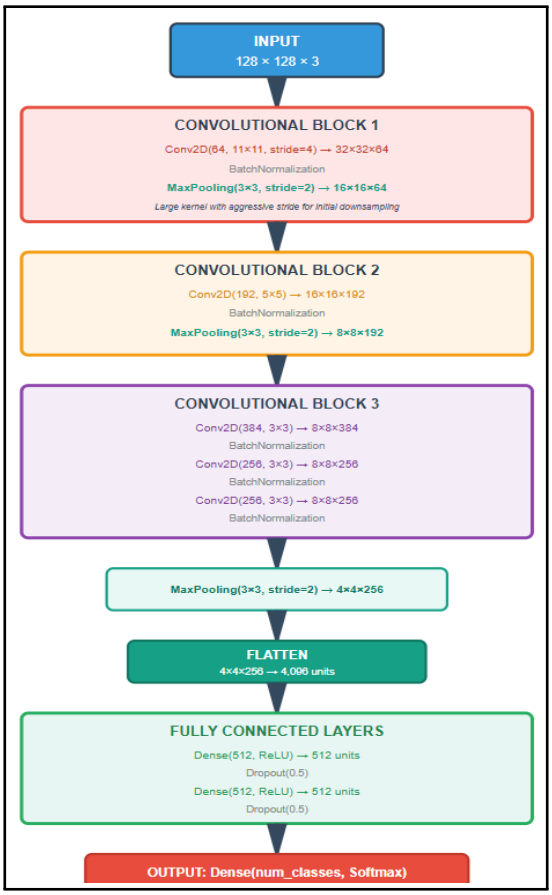


Figure 2: Modified AlexNet Architecture

B) GoogleNet

The custom GoogleNet (Inception v1) uses special Inception modules designed for image classification. The model has an input layer with two convolutional blocks. In the first block, it uses a 7x7 convolution with ReLU activation. It is then followed by a MaxPooling and BatchNormalization process. The second block consists of a 1x1 convolution and a 3x3 convolution followed by MaxPooling and BatchNormalization. After the convolution layers, the model uses Inception modules to perform parallel convolutions to capture complex patterns. The Dense layer transforms the extracted feature into a vector. Finally, the last layer uses a softmax activation function to produce the final output. Figure 3 shows the structure of the model.

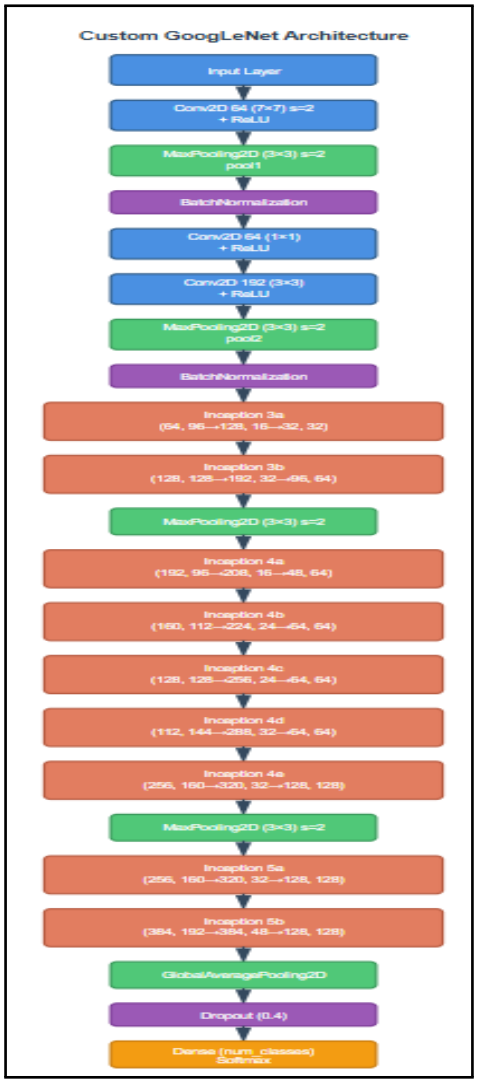


Figure 3: GoogLeNet Modified Model

C) VGG-16

The proposed VGG-16 model has an input layer that accepts images sized 224 x 224 pixels and have three channels to accept RGB images. The architecture consists of 13 convolutional layers divided into four blocks. Each of the blocks used ReLu activation, and followed by MaxPooling layers to reduce the image dimensions. The first block uses 64 filters, the second block uses 128 filters, and both the third and fourth use 256 filters. After the features are extracted, a Dropout layer is used to prevent over-fitting and a Flatten layer is used to convert the 2D feature maps into a single vector. The data is then passed through the Dense layer, which contains 1,024 neurons in the fully connected layer. Finally, the last layer contains four neurons, which produce the final output for the four classes using the Softmax activation function. Figure 4 shows the architecture of the modified VGG-16 model.

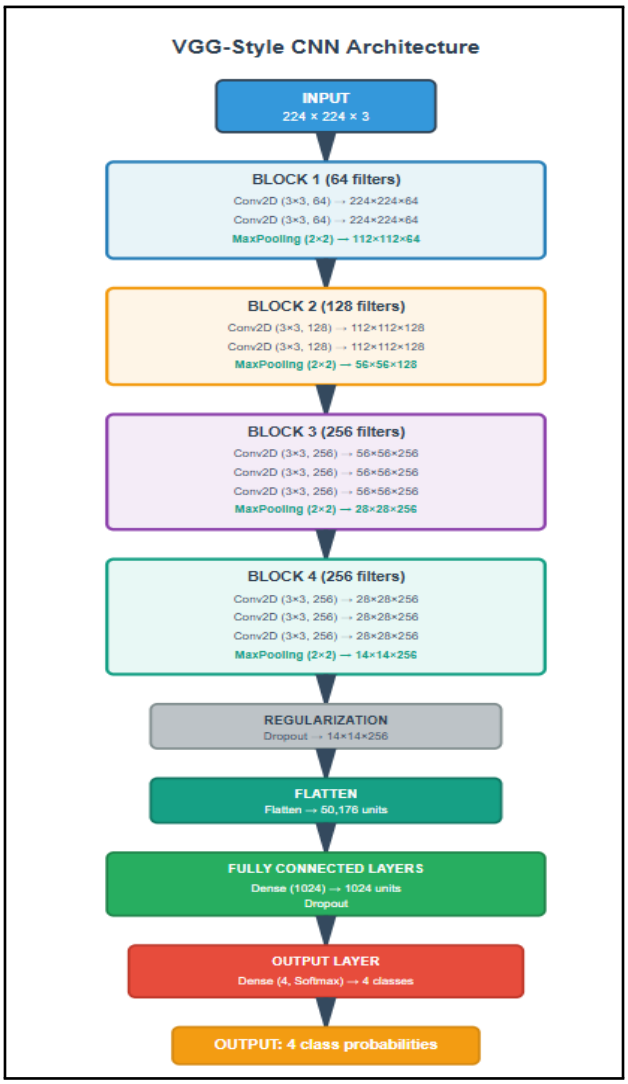


Figure 4: VGG-16 Modified Model

4 Results And Discussions

In this section, all the results acquired from our comparitive study are presented.VGG16, AlexNet and Googlenet model's evaluation is done based on several metrics such as training duration,Training vs Validation Accuracy and Loss,ROC Curves and classification reports.

4.1 Training Duration

It was very important to know the time required to train each model when it comes to potential practical implementation.So as shown in Table 4 we can say how each model differed.

Model	Duration
VGG16	6 hours 34 minutes
AlexNet	13 minutes 27 seconds
Googlenet	30 minutes 2 seconds

Table 4: Training Duration

- VGG16 model was the most complex and heavy model with alot of parameters hence it took the longest training duration of 6 hours and 34 minutes.
- The customised version of AlexNet was very efficient, since its size had been drastically reduced and it completed its training in just 13 minutes and 27 seconds.
- The modified GoogleNet also shows efficiency,trained in 30 minutes and 2 seconds.

4.2 Test Accuracy

The primary way to measure the model's performance is its accuracy on the data that was unseen, which is the unseen test set. The Table 5 gives a good summary of the final test accuracies gained for each model.

Model	Test Accuracy
VGG16	87.50%
AlexNet	84.68%
Googlenet	80.73%

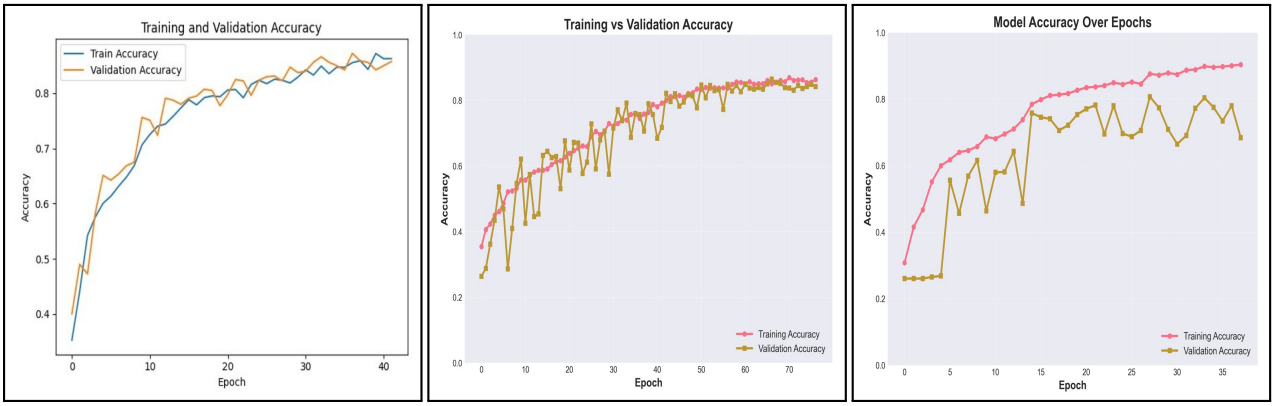
Table 5: Test Accuracy

- VGG16 was able to achieve the highest test accuracy of 87.50%.

- AlexNet model follows close to VGG16 but with less than half the size and parameters with a accuracy of 84.68% proving its efficiency.
- Last is GoogleNet which had a good accuracy of 80.73% again with less than half the size of VGG16.

4.3 Training Accuracy vs Validation Accuracy

These graphs will showcase the accuracy curves that were obtained through the training process and will give more information about how the training process went which will be crucial in helping us to solve issues like overfitting.



(a) VGG16 Training vs Validation Accuracy

(b) VGG16 Training vs Validation Accuracy

(c) VGG16 Training vs Validation Accuracy

Figure 5: Comparison of Training vs Validation Loss for VGG16, AlexNet, and GoogLeNet Models

- For all the models the curves are close to each other.This the best outcome that could come-out which means that the models are good at generalizing and our techniques like data augmentation and dropout are effective at preventing overfitting.

4.4 Training Loss vs Validation Loss

The loss curves are just as important, as they will give us the knowledge of how the loss is changing throughout the training process till the end which can help in developing better loss reduction techniques.

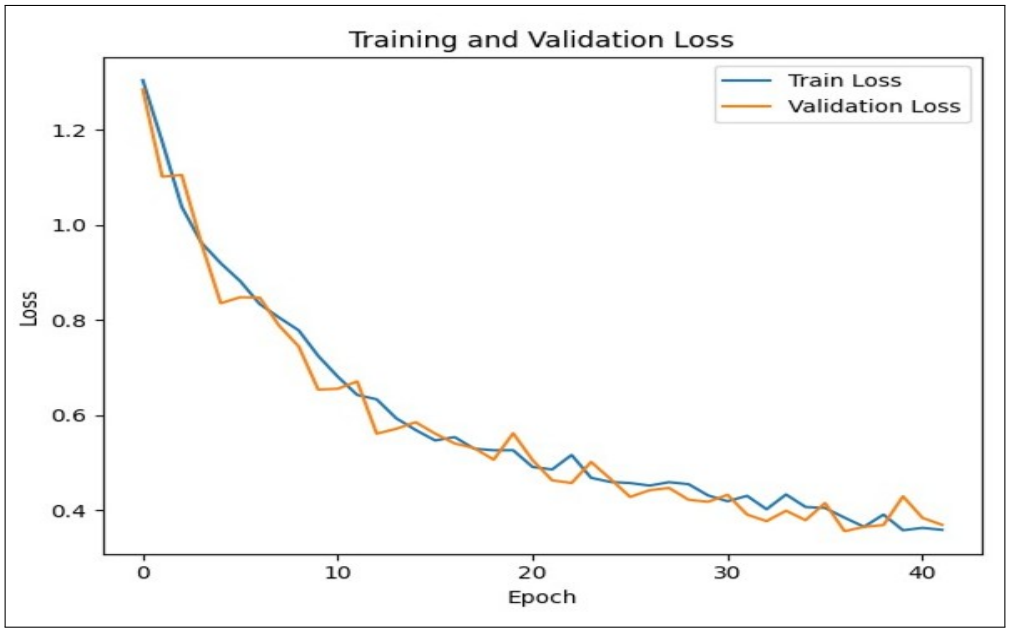


Figure 6: VGG16 Training vs Validation Loss

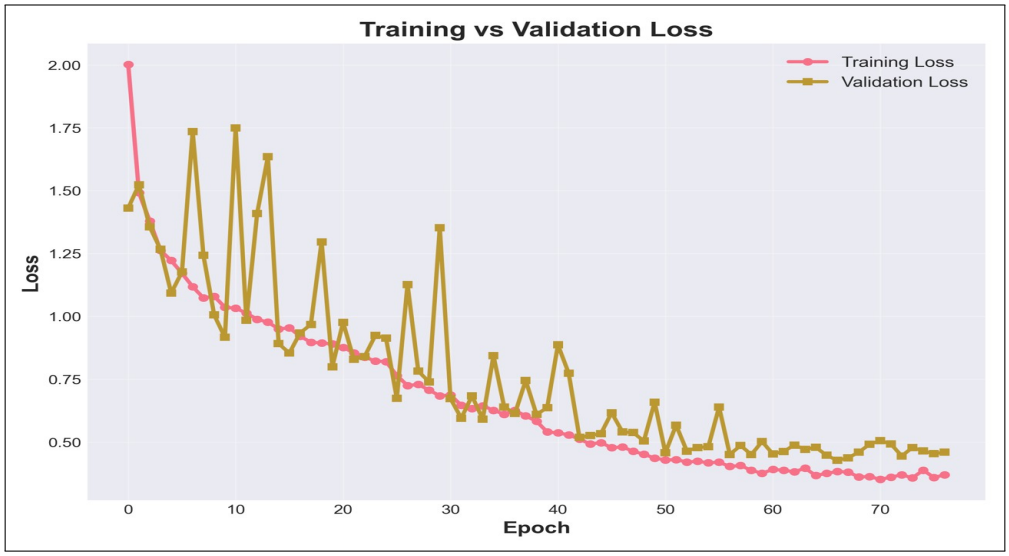


Figure 7: AlexNet Training vs Validation Loss

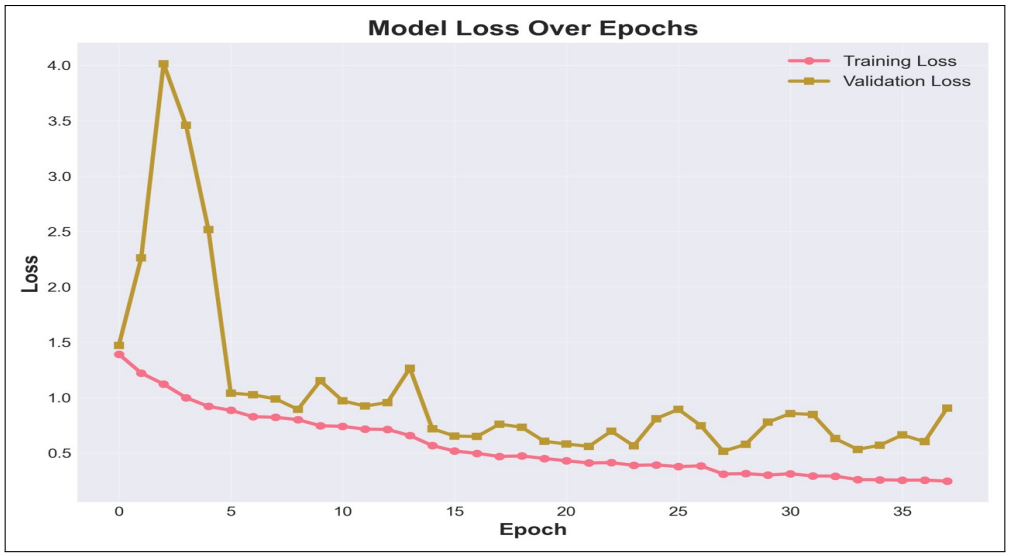


Figure 8: Googlenet Training vs Validation Loss

The consistent downward trend of the graph of the loss curves before flattening shows that the loss has decreased as training went on which is a good sign.As the two lines are close to each other, this confirms that the models did not overfit the training data.

4.5 Confusion Matrix

The confusion matrix is used to get a detailed visuals of the image classification performance each model went through. This shows all the correct predictions as well as incorrect ones.

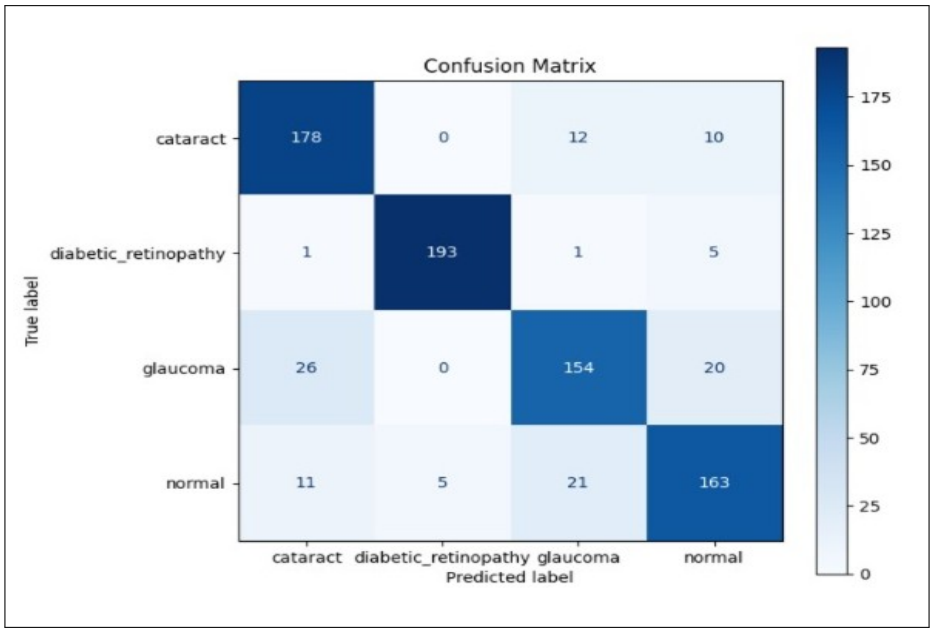


Figure 9: VGG16 Confusion Matrix

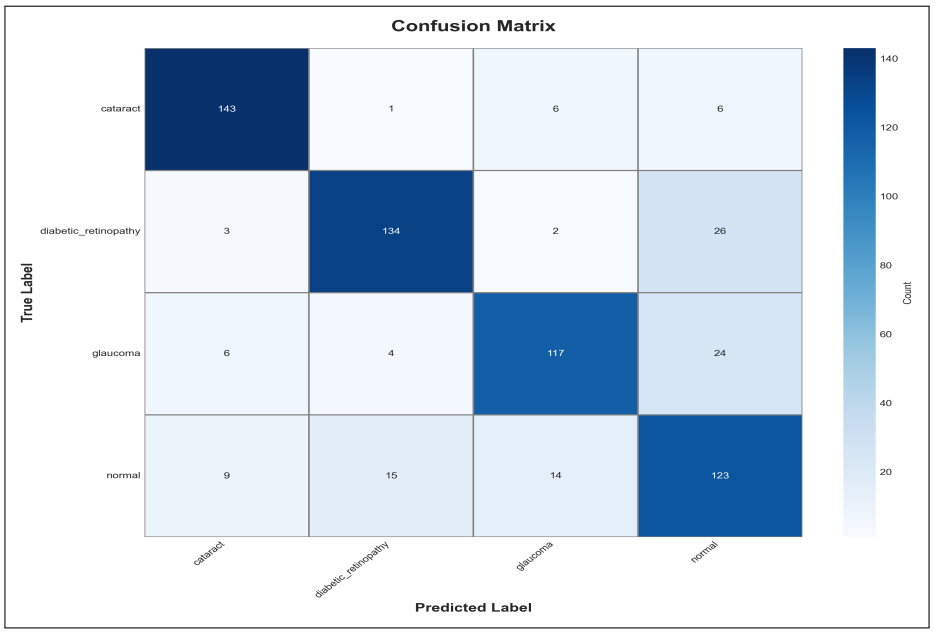


Figure 10: Googlenet Confusion Matrix

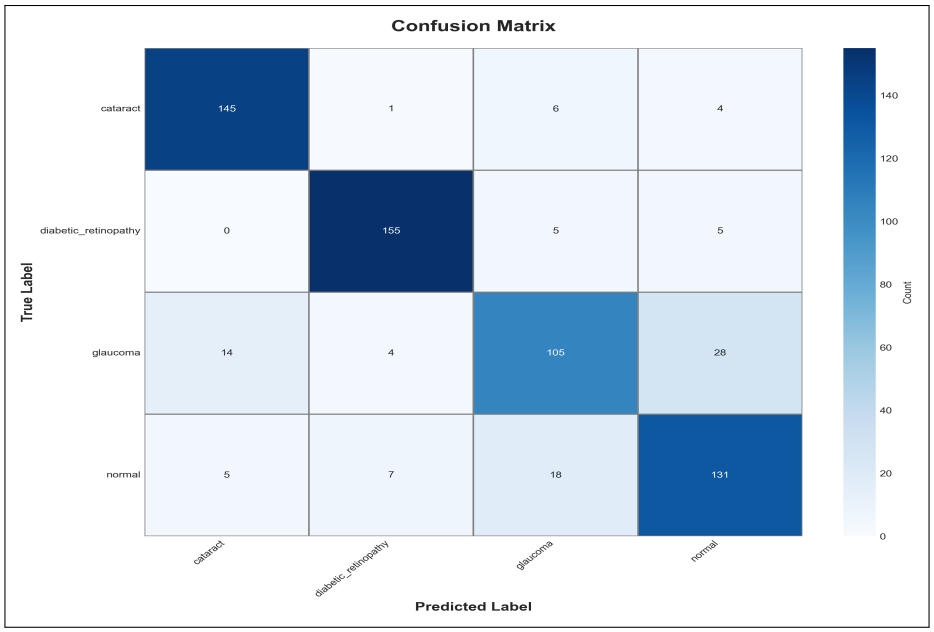


Figure 11: Alexnet Confusion Matrix

- For VGG16 9 and AlexNet 11 the models show high values along the main diagonal which confirms that the overall accuracy is high.
- For AlexNet we can observe some confusion primarily occuring between glaucoma and normal classes.
- For GoogleNet 10 the matrix is a bit more dispersed and shows pattern of errors, mainly present in high numbers in the normal class.

4.6 Classification Report

Class	Precision	Recall	F1-Score	Support
Cataract	0.82	0.89	0.86	200
Diabetic Retinopathy	0.97	0.96	0.97	200
Glaucoma	0.82	0.77	0.79	200
Normal	0.82	0.81	0.82	200

Table 6: VGG16 Classification Report

Class	Precision	Recall	F1-Score	Support
Cataract	0.8841	0.9295	0.9062	156
Diabetic Retinopathy	0.9281	0.9394	0.9337	165
Glaucoma	0.7836	0.6954	0.7368	151
Normal	0.7798	0.8137	0.7964	161

Table 7: Alexnet Classification Report

Class	Precision	Recall	F1-Score	Support
Cataract	0.8882	0.9167	0.9022	156
Diabetic Retinopathy	0.8701	0.8121	0.8401	165
Glaucoma	0.8417	0.7748	0.8069	151
Normal	0.6872	0.7640	0.7235	161

Table 8: Googlenet Classification Report

4.7 ROC Curve

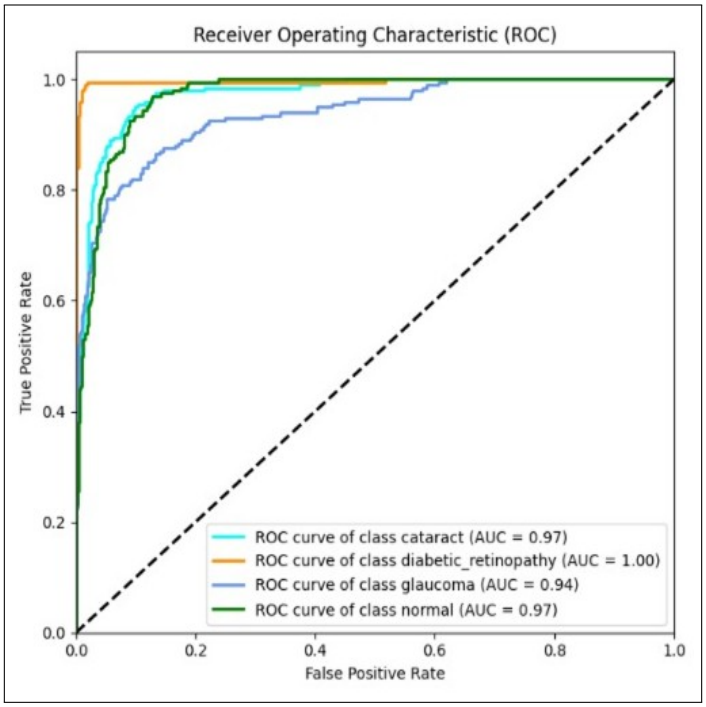


Figure 12: VGG16 ROC Curve

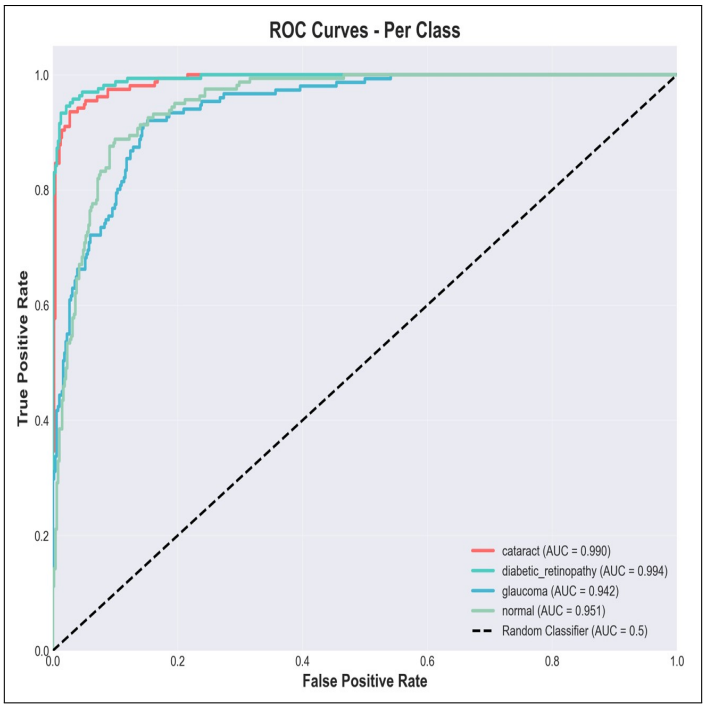


Figure 13: Alexnet ROC Curve

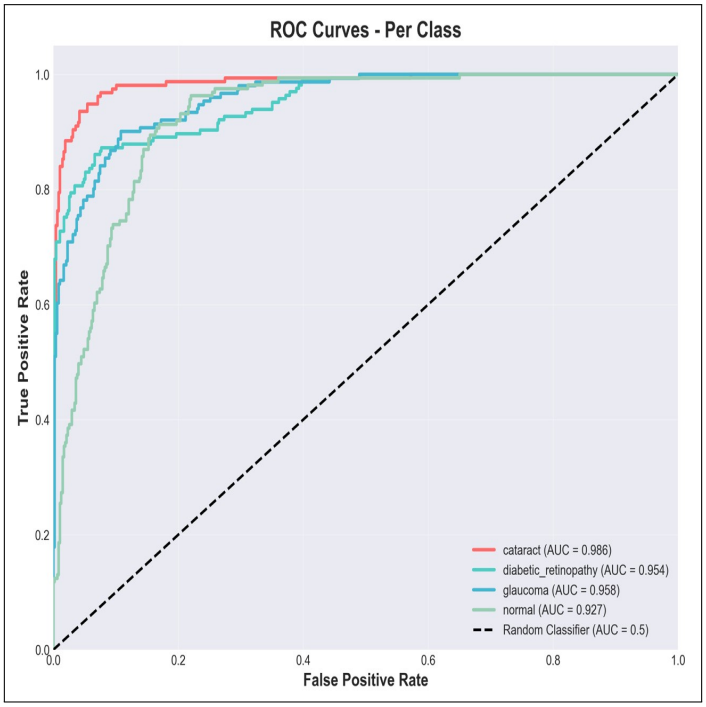


Figure 14: Googlenet ROC Curve

- All the models show good classification ability where their ROC curves are going sharply towards the top-left corner and achieving high AUC scores.

- The AUC scroes for VGG16 and AlexNet are high for the Cataract and Diabetic Retinopathy classes nearing 0.98 which is almost near perfect separation.

4.8 Final Predictions

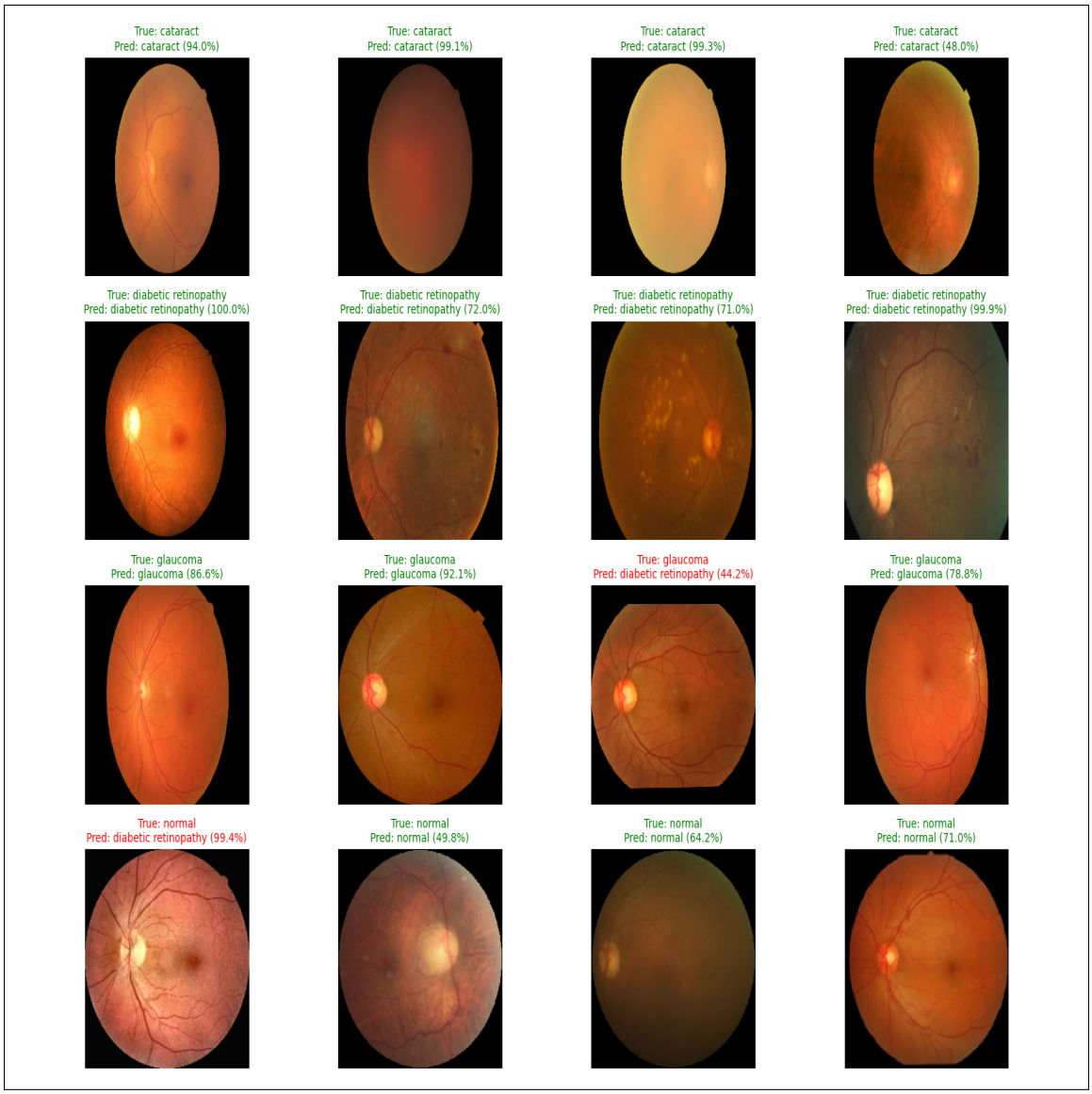


Figure 15: VGG16 Predictions

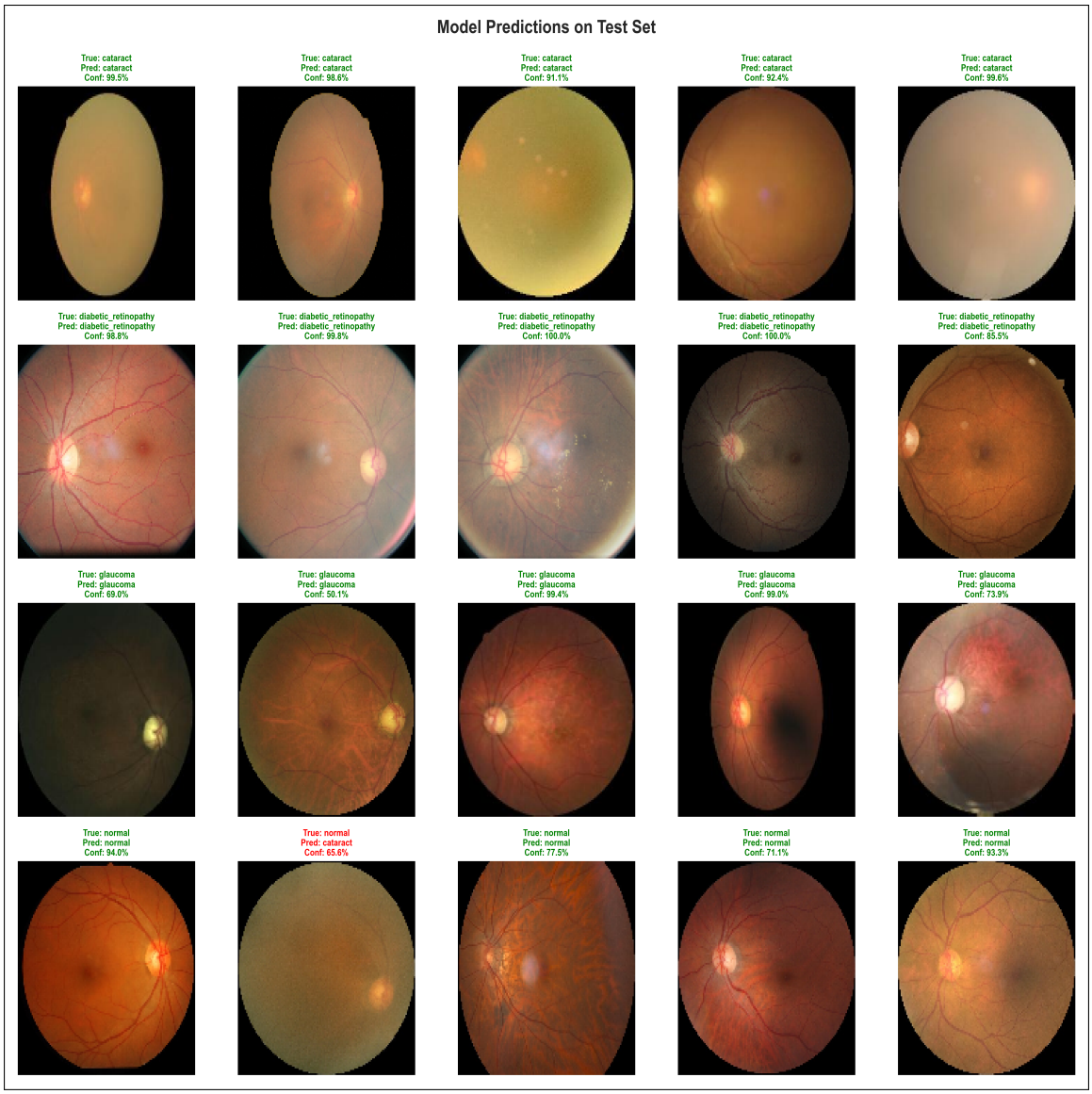


Figure 16: Alexnet Predictions

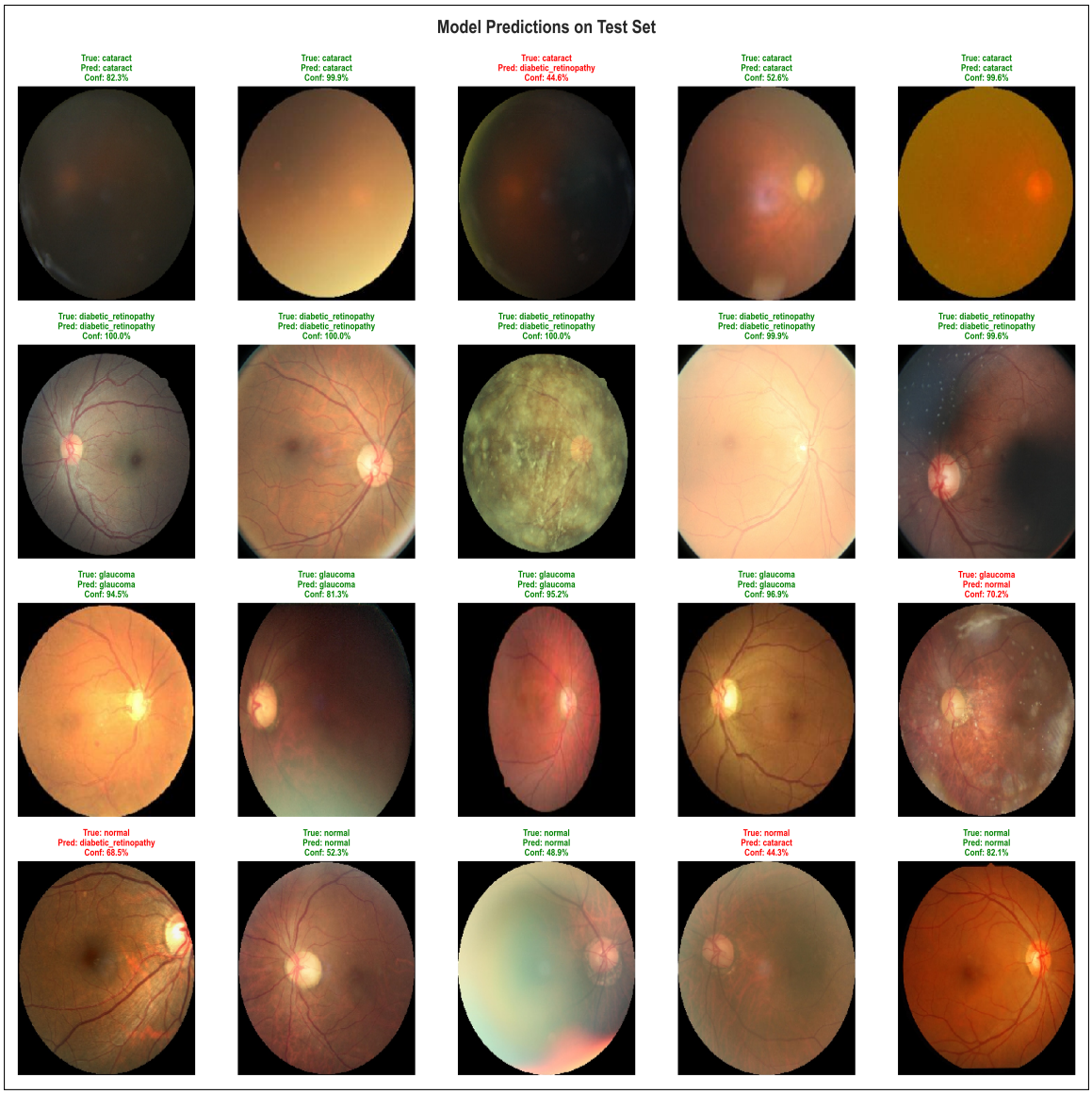


Figure 17: Googlenet Predictions

5 Conclusion

Eye disease detection is necessary for early diagnosis and prevention of vision loss. The system can detect ocular conditions automatically using deep learning methods with CNN architectures. The healthcare industry will save much time on diagnostic screening by detecting eye diseases automatically. This study implemented three CNN architectures: VGG-16, AlexNet, and GoogleNet for the classification of four classes namely Cataract, Diabetic Retinopathy, Glaucoma, and Normal.

This study uses VGG-16 Architecture gets the best results when using Adam optimizer, learning rate 0.0001, batch size 32, and epoch 50. The VGG-16 model obtained the highest accuracy rate of 87.50% with precision values ranging from 0.82 to 0.97 across different classes. AlexNet achieved competitive performance with an accuracy of 84.68% and significantly faster training time of 13 minutes 27 seconds compared to VGG-16's 6 hours 34 minutes, making it suitable for resource-constrained environments. GoogleNet demonstrated balanced performance with 80.73% accuracy and moderate training time of 30 minutes 2 seconds. The experimental results show that all three models can effectively classify eye diseases with VGG-16 providing the best diagnostic accuracy, particularly achieving 97% precision for Diabetic Retinopathy detection.

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